# Loops

## There are two kinds of **while** loops. What are they?

While loop
Do-while loop
Which of these has the condition at the end? Do-while loop
Which of these requires a semi-colon? Do-while loop
Which of these iterates a max of zero times?while loop
Which of these iterates a min of 1 time?Do-while loop
Which is referred to as a precondition loop?while loop
Which of these is referred to as a postcondition loop?Do-while loop
Describe a problem that can be solved using a loop
Asking a user a math question and keep asking until they get the right answer.
While loops iterate over and over again as long as its condition is __ture_____?
Describe why you would use a post condition loop over a pre condition loop.
If you need the code to execute once before checking the condition

## Up and Down

Print the numbers 1 to 10 and then 10 to 1 using two while loops.

Example:

```
1 2 3 4 6 7 8 9 10 10 9 8 7 6 5 4 3 2 1
```

```
let num=1;

while(num<=_10__)

{

    console.log(num);

    num++;

}

num--;

while(num>=__1_)

{
```

```
    console.log(num);
    num--;
}
```

## Product of 5

Use a while loop to calculate the product of all numbers from 1 to 5.

```
let num=1;
let product=1;
while(num<=_5_)
{
        product=product*_num__;
        num++;
}
console.log("The product of all numbers from 1 to 5 is " + __product_____);
```

## Sum of Cubes

Ask a user for a number and display the sum of all cubes from 1 to the number.

```
num = int(input("Enter a number: "))
sum_cubes = 0
for i in range(1, num + 1):
    sum_cubes += i ** 3
print("Sum of cubes:", sum_cubes)
```

Example:

```
User enters 3 and you display 14.
```

## Ask Until 0

Use a while loop to keep asking a user for a number until the number entered is 0.

```
while True:
    n = int(input("Enter a number (0 to stop): "))
    if n == 0:
        print("Stopped.")
        break
```

## Count by 1

In this exercise, you are going to count by 1, starting at 1 and ending with the provided number (ask the user for thai number).

```
end = int(input("Enter a number: "))

for i in range(1, end + 1):

    print(i, end=" ")

print()
```

Example:

```
The user enters 4 and you display:

1 2 3 4
```

# Down by 1

In this exercise, you are going to count by 1, starting at a number the user enters and ending with 1.

```
start = int(input("Enter a starting number: "))

for i in range(start, 0, -1):

    print(i, end=" ")

print()
```

Example:

```
The user enters 5 and you display:


5 4 3 2 1
```

## Count by 2

In this exercise, you are going to count by 2, starting at 2 and ending with the provided number (ask the user for thai number).

```
end = int(input("Enter a number: "))

for i in range(2, end + 1, 2):

    print(i, end=" ")

print()
```

Example:

```
The user enters 10 and you display:


2 4 6 8 10
```

## How many times?

In this exercise, you ask for a starting value, an ending value, and an increment. Display how many times the loop executes given these parameters.

```
start = int(input("Enter a starting value: "))
```

```
end = int(input("Enter an ending value: "))

step = int(input("Enter an increment: "))



count = 0

for i in range(start, end + 1, step):

    count += 1

print("The loop executed", count, "times.")
```

Example:

```
The user enter 3, 10 and 3 and you then display 3

The user enter 5, 15 and 1 and you display 11
```

# Countdown

For this exercise, you are going to return an output that counts down from a provided starting value. Each number should be space separated.

```
start = int(input("Enter a starting value: "))

for i in range(start, -1, -1):

    print(i, end=" ")

print()
```

Example:

```
The user enters 10 and you display 10 9 8 7 6 5 4 3 2 1 0
```

```
The user enters 5 and you display 5 4 3 2 1 0
```

# Sum Numbers

Given a starting value, ending value, and an increment value, add up all the numbers in the range (inclusive of the end points).

```
start = int(input("Enter a starting value: "))

end = int(input("Enter an ending value: "))

step = int(input("Enter an increment value: "))



total = 0

for i in range(start, end + 1, step):

    total += i

print("Sum =", total)
```

Example:

```
The user enters 0, 10 and 2 and you displays 30
```

```
The user enters 25, 30 and 5 and you display 55
```

# Average

In this exercise, you are going to return the average over a range of numbers, from the start value to the end value (inclusive of both endpoints).

```
start = int(input("Enter a start value: "))

end = int(input("Enter an end value: "))



count = 0

total = 0

for i in range(start, end + 1):

    total += i

    count += 1



average = total / count

print("Average =", average)
```

Example:

```
The user enters 10 and 20 and you display 15

The user enters 0 and 1000 and you display 500
```

# Factorial

In this exercise, you are given a number. You should print the factorial of the number. Remember factorial is multiplying that number times all of the numbers below it.

5 factorial = 5 * 4 * 3 * 2 * 1

```
num = int(input("Enter a number: "))

fact = 1

for i in range(1, num + 1):

    fact *= i

print("Factorial =", fact)
```

Example:

```
The user enters 5 and you display 120

The user enters 3 and you display 6
```

# Exercise : Modified count by 2

In this exercise, you are going to count by 2s starting at 2 and ending at the provided number. You should skip any number that ends in a 0.

Print the output as a string.

```
end = int(input("Enter a number: "))

output = ""
```

```
for i in range(2, end + 1, 2):
    if i % 10 != 0:
        output += str(i) + " "

print(output.strip())
```

Example:

```
The user enter 14 and you display 2 4 6 8 12 14

The user enters 10 and you display 2 4 6 8
```